Parul University
Faculty of Engineering and Technology
Department of Applied Science & Humanities
Academic Year 2026-27
Subject: Quant and Reasoning (303105311)
Branch: CSE/ IT

TUTORIAL-6

21. How many 4-letter words with or without meaning, can be formed out of the letters of the word, 'LOGARITHMS', if repetition of letters is not allowed?

(a) 40 (b)400 (c) 5040 (d) 2520

22. Three boys and Three girls are to be seated around a table in a circle. Among the boys, X does not want any girl adjacent to him and the girl Y does not want any boy adjacent to her. How many such arrangements are possible ?

(a) 8 (b)2 (c) 4 (d) 6

23. In how many ways can 10 identical presents be distributed among 6 children so that each child gets atleast one present ?

(a) $^{15}C_5$ (b) $^{16}C_6$ (c) $^{9}C_5$ (d) 6^{10}

24. A set of 15 different words are given. In how many ways is it possible to choose a subset of not more than 5 words ?

(a) 4944 (b) 4^{15} (c) 15^4 (d) 4943

25. How many distinct words can be formed out of the word PROWLING which start with R and end with W ?

(a) 8!/2! (b) (c) 15^4 (d) 4943

26. There is a question paper consisting of 15 questions. Each question has an internal choice of 2 options. In how many ways can a student attempt one or more questions of the given 15 questions in the paper?

(a) 3^7 (b) 3^8 (c) 3^{15} (d) 3^{15}-1

27. In how many ways is it possible to choose a white square and a black square on a chess board so that the squares must not lie on the same row or column?

(a) 56 (b) 896 (c) 60 (d) 768

28. Find the sum of number of sides and number of diagonals of a hexagon.

(a) 210 (b) 15 (c) 6 (d) 9

29. There is a number lock with 4 rings. How many attempts at the maximum would have to be made before getting the right number ?

(a) 10^4 (b) 255 (c) 10^4-1 (d) None of these

30. There are 4 letters and 4 envelopes. In how many ways can wrong choices be made ?

(a) 4^3 (b) 4!-1 (c) 16 (d) 4^4-1

31. There is a 7-digit telephone number with all different digits. If the digit at extreme right and extreme left are 5 & 6 respectively, Find how many such telephone numbers are possible ?

(a) 120 (b) 100000 (c) 6720 (d) None of these

32. If we have to make 7 boys sit alternately with 7 girls around a round table which is numbered, then the number of ways in which this can be done is

(a) $2x(7!)^2$ (b) 7! X 6! (c) 7! X 7! (d) None of these

33. Seven different objects must be divided among three people. In how many ways can this be done if one or two of them can get no objects ?

(a) 15120 (b) 2187 (c) 3003 (d) 792

34. In how many ways a cricketer can score 200 runs with fours and sixes only ?

(a) 13 (b) 17 (c) 19 (d) 16

35. A dice is rolled six times. 1,2,3,4,5,6 appears on consecutive throws of dice. How many ways are possible of having 1 before 6?

(a) 120 (b) 360 (c) 240 (d) 280

36. There are 20 people among whom two are sisters. Find the number of ways in which we can arrange them around a circle so that there is exactly one person between the two sisters?

(a) 18! (b) 2!.19! (c) 19! (d) None of these

37. How many rectangles can be formed out of a chessboard ?

(a) 204 (b) 2!.19! (c) 19! (d) None of these

38. How many 4-digit numbers that are divisible by 4 can be formed from the digits 1,2,3,4,5?

(a) 36 (b) 72 (c) 24 (d) None of these

39. There are 5 bottles of sherry and each have their respective caps. If you are asked to put the correct cap to the correct bottle then how many ways are there so that not a single cap is on the correct bottle ?

(a) 44 (b) 5^5-1 (c) 5^5 (d) None of these

40. How many 6-digit numbers contain exactly 4 different digits ?

(a) 4536 (b) 294840 (c) 191520 (d) None of these